MILITARY SPECIFICATION

OSCILLOSCOPE (AN/USM-425(V)), 100 MEGAHERTZ (MHZ), GENERAL PURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solid state, general purpose, dual trace oscilloscopes with a delayed time base and X-Y capability through the vertical channels. Hereinafter, the oscilloscopes are called equipment.

1.2 Classification. Equipment covered by this specification may be commercially available oscilloscopes modified to the extent necessary to meet the following descriptions.

1.2.1 Configuration A. Configuration A equipment will be as defined in MIL-T-28800 and herein for Type II, Class 2, Type A, and color forest green. Alternate colors may be specified in the contract or order.

1.2.2 Configuration B (AN/USM-425(V)1). Configuration B equipment will be as defined in MIL-T-28800 and herein for Type II, Class 4, Style C, and color-optional with the manufacturer.

1.2.3 Configuration C. Configuration C equipment will be as defined in MIL-T-28800 and herein for Type II, Class 4, Style E, and color-optional with the manufacturer.

1.2.4 Confiugration D. Configuration D equipment will be portable and as defined in MIL-P-28800 and herein for Type III, Class 5, Style E, and color optional with the manufacturer.

2.2 APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-C-28777	Cable Assembly, Electronic Test Equipment (3 Wire, 3 Prong, 125 Volt, Molded) Grounding Plug Connector
MIL-T-28800	Test Equipment For Use With Electrical And Electronic Equipment, General Specification For
MIL-C-83286	Costing Urethane, Aliphatic Isocyanate, For Aerospace Applications
MIL-P-83789	Power Supply, Portable Battery Pack, Pp- /U

STANDARDS

Federal

FED-STD-595	Colors

Military

MS3102	Connector, Receptacle, Electric, Straight, Solder Contacts, AN Type
MS3106	Connector, Plug, Electric, Straight, Solder Contacts, AN Type
MS22435	Power Receptacle, Aircraft, Electronic Test Equipment, Wiring Of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Equipment configurations. Unless otherwise specified, requirements of this specification shall apply to all configurations (See 6.2.1.2).

3.1.1 Configuration A. The equipment shall be as required in MIL-T-28800 and herein for Type II, Class 2, Style A, and color as specified herein.

3.1.2 Configuration B. The equipment shall be as required in MIL-T-28800 and herein for Type II, Class 4, Style C, and color R. The equipment shall be portable.

3.1.3 Configuration C. The equipment shall be as required in MIL-T-28800 and herein for Type II, Class 4, Style E, and color R. The equipment shall be convertible/rackmountable.

3.1.4 Configuration D. The equipment shall be portable and as required in MIL-T-28800 and herein for Type III, Class 5, Style E, and color R.

3.2 Safety. The equipment shall comply with the Type III safety requirements of MIL-T-28800.

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3.3 Parts, materials, and processes. Unless otherwise specified herein, the equipment shall comply with the Type III parts, materials, and processes of MIL-T-28800.

3.3.1 Restricted materials. The equipment shall comply with the restricted material requirements of MIL-T-28800 including the requirement for mercury or radioactive materials; and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3.3.2 Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be Type BNC female (See 3.13.5).

3.3.3 Configuration A color. The equipment enclosure shall be finished forest green, 24052 of FED-STD-595 in accordance with MIL-C-83286. The front panel shall be color T of MIL-T-28800. Alternate colors may be specified in contract or order (See 6.2.1.3).

3.4 Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type III design and construction requirements of MIL-T-28800.

3.4.1 Solid state construction. Except for the cathode ray tube (CRT), electron tubes shall not be used. The equipment shall be of solid state, modular miniaturized construction.

3.4.2 Mainframe plug-in concept. Equipment utilizing externally accessible, externally removable subassemblies (mainframe plug-in concept) do not meet the requirements of this specification and therefore are unacceptable. Items using this concept may be offered, provided the plug-in(s) shall not be removable by access through the front panel or rear panel with the unaided hand. A system part number shall be assigned to any such mainframe plug-in combination required to meet the requirements of this specification. This system part number shall be marked on the mainframe identification plate and any plug-in(s).

3.4.3 Controls. Unless otherwise specified herein, built-in adjustments and compensating devices shall not be externally accessible. All potentiometer controls shall have at least 270° rotation range.

3.4.4 Accessibility. The design of the equipment shall provide accessibility in accordance with a through c.

a. Subassemblies and chassis components can be removed without removing any other hard wired printed circuit card, subassembly or component.

b. Adjustments can be made without removing any component, printed circuit card or subassembly except the use of extender cards is permitted.

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c. Printed circuit cards can be removed without the need to unsolder cables and interconnecting wiring (connections to all printed circuit cards shall be through pin and socket connectors). Printed circuit cards (mother boards) designed primarily to distribute power and signals to other printed cards (daughter boards) are excluded from this requirement. When such mother boards are used, they shall be accessible from both sides.

3.4.5 Encapsulation and embedment. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits.

3.5 Electrical power sources and connections.

3.5.1 Input power consumption. The equipment shall consume not more than 100 volt-amperes.

3.5.2 Configuration A power sources and connections. The equipment shall comply with the Type I electrical power sources and connections requirements for Style A of MIL-T-28800 and as specified herein.

3.5.2.1 Configuration A power sources. The equipment shall operate from the Type I nominal and alternate power sources of MIL-T-28800 in accordance with a and b. Internal battery capability shall not be provided.

a. 50, 60, and 400 Hertz (Hz), single phase 115/230 Volts.

b. DC external battery pack (compatible with the battery pack specified in MIL-P-83789).

3.5.2.1.1 Configuration A electrical power transients. The equipment shall operate under voltage and frequency transient conditions as specified for Type III of MIL-T-28800.

3.5.2.2 Configuration A electrical power connections. Unless otherwise specified in the contract or order the electrical power connection shall be in accordance with MIL-T-28800 for Style A enclosures as specified for 400 Hz single phase 115 Volts alternating current (VAC) operation and external direct current (DC) (not both at the same time). The MS3102R-18-9P mating receptacle shall be wired in accordance with MS24435 except pin D shall not be internally grounded. Voltage shall not be coupled to unused power connections. The power cables shall be in accordance with a through d. Unless otherwise specified in the contract or order, only the 115 VAC power cable shall be supplied.

a. For 115 VAC, 230 VAC and external DC operation, the equipment end of the power cable shall terminate with an MS3106R-18-9S connector. The black lead shall be attached to pin B, the white lead shall be attached to pin D and the green lead shall be attached to pin G.

b. For 115 VAC operation, the source end of the power cable shall comply with the plug connector requirements of MIL-C-28777.

c. For 230 VAC operation, the source end of the power cable shall be as specified in the contract or order (See 6.2.1.4).

d. For external DC operation, the source end of the power cable shall terminate with an MS3106R-18-9P connector.

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3.5.2.2.1 Configuration A power selection switch. A power selection switch shall be provided as specified for Type III of MIL-T-28800 (external battery, 115 VAC, 230 VAC) (See 3.5.2.3).

3.5.2.3 Configuration A fuses and fuseholders. One side of the input power lines shall be fused. Either a common or separate fuseholders shall be provided for 115 VAC and 230 VAC operation. If only one fuseholder is used (common), the 230 VAC fuse shall be stored in the front panel. The equipment shall be delivered with the 115 VAC fuse installed (if a common fuseholder is utilized) and with the power selection switch in the 115 VAC position. The design of the fuseholder(s) shall not permit finger contact with live conductors.

3.5.3 Configuration B, C and D power sources and connections. The equipment shall comply with the Type III electrical power sources and connections requirements of MIL-T-28800 and as specified herein.

3.5.3.1 Configuration B, C, and D power sources. The equipment shall operate from the Type III nominal and alternate power source of MIL-T-28800 for 50, 60, and 400 Hz single phase 115/230 Volts. Internal battery capability shall not be provided.

3.5.3.1.1 Configuration B, C, and D electrical power transients. The equipment shall operate under voltage and frequency transient conditions as specified for Type III of MIL-T-28800.

3.5.3.2 Configuration B, C, and D electrical power connections. The equipment electrical power connections shall be as specified in MIL-T-28800 for Type III except the capability for separate 115 VAC and 230 VAC power cords shall be provided. Voltage shall not be coupled to unused power connections. The power cables shall be in accordance with a and b. Unless otherwise specified in the contract or order, only the 115 VAC power cord shall be supplied.

a. For 115 VAC operation, the power cable shall be in accordance with MIL-C-28777/3 except that the molded connector at the equipment end shall mate with Switchcraft Part Number EAC-302 connector, or equal.

b. For 230 VAC operation, the power cable shall be as specified in the contract or order (See 6.2.1.4).

3.5.3.2.1 Configuration B, C, and D input power switch. An input power switch shall be provided which shall break all poles of the source except the safety grounding conductor when in the off position.

3.5.3.3 Configuration B, C, and D fuses and fuseholders. One side of the input power line shall be fused. Either a common or separate fuseholders shall be provided for 115 VAC and 230 VAC operation. If only one fuseholder is used (common), the 230 VAC fuse shall be stored in the front panel cover or otherwise externally attached to the equipment's rear panel. The equipment shall be delivered with the 115 VAC fuse installed (if a common fuse holder is utilized) and with the power selection switch in the 115 VAC position. The design of the fuseholder shall not permit finger contact with live conductors.

3.5.3.4 Configuration B, C, and power selection switch. A power selection switch shall be provided as specified for Type III in MIL-T-28800 (115 VAC, 230 VAC).

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3.6 Dimensions, weight and mechanical stability. The equipment shall comply with the dimensions, weight, and mechanical stability requirements of MIL-T-28800 and as specified herein.

3.6.1 Configuration A weight and dimensions.

3.6.1.1 Configuration A weight. The maximum weight of the equipment without the front cover and accessories shall not exceed 15 kg (33.1 pounds). The maximum weight of the equipment with the front cover and all accessories in place shall not exceed 18 kg (39.7 pounds).

3.6.1.2 Configuration A dimensions. The maximum dimensions of the equipment with the front cover and all accessories in place shall not exceed 356 mm (14.0 inches) in width, 267 mm (10.5 inches) in height, and 597 mm (23.5 inches) in depth.

3.6.2 Configuration B weight and dimensions.

3.6.2.1 Configuration B weight. The maximum weight of the equipment without the front cover and accessories shall not exceed 14.5 kg (32 pounds). The maximum weight of the equipment with the front cover and all accessories in place shall not exceed 17 kg (37.5 pounds).

3.6.2.2 Configuration B dimensions. Partial dimensions of the equipment, excluding the tilt bail handle and the front cover shall not exceed 343 mm (13.5 inches) in width, 172 mm (6.75 inches) in height, and 597 mm (23.5 inches) in depth. The frontal area shall not exceed 591 cm^2 (91.5 $inches^2$). Overall dimensions of the equipment shall not exceed 375 mm (14.75 inches) in width, 197 mm (7.75 inches) in height, and 648 mm (25.5 inches) in depth. The tilt bail handle assembly shall not increase the width more than 50 mm (2.0 inches) and the depth by more than 75 mm (3.0 inches) in carrying position (See 3.11.2.2). Clearance between the front cover and the grip portion of the handle shall be no less than 50 mm (2.0 inches) at its closest proximity.

3.6.3 Configuration C weight and dimensions.

3.6.3.1 Configuration C weight. The maximum weight of the equipment shall not exceed 15 kg (33.1 pounds).

3.6.3.2 Configuration C dimensions. Partial dimensions of the equipment behind the front panel shall not exceed 597 mm (23.5 inches) in depth, 130 mm (5.1 inches) in height, and 442 mm (17.6 inches) in width including slides. Nominal size of the front panel when converted for rack mounting shall be 133.4 mm (5-1/4 inches) high and in accordance with the rack mounting requirements of MIL-T-28800.

3.6.4 Configuration D weight and dimensions.

3.6.4.1 Configuration D weight. The maximum weight of the equipment shall not exceed 10 kg (22 pounds).

3.6.4.2 Configuration D dimensions. Overall dimensions of the equipment shall not exceed 312 mm (12.3 inches) in width, 255 mm (10.0 inches) in height, and 535 mm (21.1 inches) in depth. The frontal area shall not exceed 552 cm^2 (85.6 $inches^2$). The addition of a tilt bail handle assembly shall not increase

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the width by more than 50 mm (2.0 inches) and the depth by more than 75 mm (3.0 inches) in the carrying position (See 3.11.2.2). Clearance between the front of the instrument and the grip portion of the handle shall be no less than 50 mm (2.0 inches) at its closest proximity.

3.7.1 Configuration A, B, and D operating attitudes. The enclosures shall be designed to facilitate horizontal viewing for bench use, and vertical viewing when the oscilloscope is placed on the floor or deck.

3.7.2 Configuration A enclosures. The enclosures shall comply with the Style A requirements of MIL-T-28800.

3.7.3 Configuration B enclosures. The enclosures shall comply with the Style C requirements of MIL-T-28800, except a removable tilt bail handle shall be supplied in lieu of the Style C handle specified in MIL-T-28800. The handle shall permit the equipment to be carried face up with the longest dimension vertical and shall tilt and support the equipment for easier viewing. The design of the handle requires procuring activity approval (See 6.2.2.1).

3.7.4 Configuration C enclosures. The enclosures shall be as specified in MIL-T-28800 for Style E and shall be convertible/rackmountable. The equipment shall be configured for rackmounting and shall include all necessary hardware including any filler panels required to conform to 3.6.3.2.

3.7.4.1 Configuration C rackmount slides. Rackmount slides shall be supplied (but not necessarily installed) with the equipment to permit access to the rear panel of the equipment when rackmounted and in the extended position.

3.7.5 Configuration D enclosures. The enclosures shall comply with the Style E requirements of MIL-T-28800. A tilt bail shall be provided to tilt and support the equipment longest dimension , excluding depth, oriented vertically. The fixed handle shall have a grip portion at least 10 cm (4 inches) long with at least 3.8 cm (1.5 inches) clearance between the handle and the case. When a tilt bail handle assembly is provided to meet this requirement, the handle requires procuring activity approval (See 6.2.2.1).

3.8 Marking and identification.

3.8.1 Configuration A marking and identification. The equipment shall comply with the Type III marking and identification requirements of MIL-T-28800 and as specified herein (See 6.2.2.2, 6.2.2.3).

3.8.1.1 Identification plate. The identification plate shall be in accordance with Type I of MIL-T-28800. The physical dimensions of the identification plate shall be such as to fit the equipment front panel.

3.8.1.2 Marking for accessories. The inside of the front panel cover shall be marked in accordance with MIL-T-28800, except it shall include separate lists for operating and nonoperating accessories. Markings on operating and nonoperating accessories shall be:

a. Stamped, etched or engraved on the accessories, or

b. Stamped, etched or engraved on durable, nonmetallic bands or tags which are securely attached to the accessories.

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3.8.1.3 Engineering design changes. Identification marking of engineering design changes shall be in accordance with the Type I requirement of MIL-T-28800 (See 6.2.2.4).

3.8.1.4 Enclosure marking. The volume, weight, and type designation of the AN nomenclature shall be marked on the exterior of the enclosure in accordance with the Type I requirements of MIL-T-28800.

3.8.2 Configuration B, C, and D marking and identification. Except for the identification plate and enclosure marking of Configuration B and C, the marking and identification B, C, and D shall comply with the Type III marking and identification requirements of MIL-T-28800. For Configuration B anc C, the identification plate shall be as specified in 3.8.1.1; the enclosure marking shall be as specified in 3.8.1.4.

3.9 Environmental requirements.

3.9.1 Configuration A environmental requirements. The equipment shall comply with the Class 2, Style A environmental requirements of MIL-T-28800 and as specified herein.

3.9.1.1 Configuration A salt atmosphere. The equipment shall comply with the Style A structural parts and enclosure salt atmosphere exposure requirements of MIL-T-28800.

3.9.2 Configuration B environmental requirements. The equipment shall comply with the Class 4, Style C environmental requirements of MIL-T-28800 and as specified herein.

3.9.2.1 Configuration B salt atmosphere. The equipment shall comply with the Style C structural parts salt atmosphere exposure requirements of MIL-T-28800.

3.9.3 Configuration C environmental requirements. The equipment shall comply with the Class 4, Style E environmental requirements of MIL-T-28800 and as specified herein.

3.9.3.1 Configuration C salt atmosphere. The equipment shall comply with the Style E structural parts salt atmosphere exposure requirements of MIL-T-28800.

3.9.4 Configuration D environmental requirements. The equipment shall comply with the Class 5, Style E environmental requirements of MIL-T-28800.

3.9.5 Electromagnetic compatibility. The equipment shall meet the requirements of MIL-T-28800 except as specified herein.

3.9.5.1 Electromagnetic interference (EMI). The equipment shall comply with the EMI requirements of MIL-T-28800 as specified in a through i.

a. CE 01: Narrow band requirement from 10 kilohertz (kHz) to 20 kHz only.

b. CE 03: Full limits.

c. CS 01: Full limits.

d. CS 02: Full limits.

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e. CS 06: Full limits.

f. RE 01: Relaxed 10 decibels (dB) at the fundamental, third harmonic, and fifth harmonic of the power source freqeuency.

g. RE 02: Full limits to ten times the highest used or intentionally generated frequency or 1 Gigahertz (GHz), whichever is greater.

h. RS 01: Full limits.

i. RS 03: Full limits to 1 GHz.

3.9.5.2 Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800 except any trace distortion (blowing, trace rotation) shall not exceed one minor division in the presence of a 5 Oersted magnetic field when the flux lines of the magnetic field and the CRT electron beam are oriented at right angles. Damage to the equipment shall not occur in the presence of up to a 20 Oersted field. Normal operation of the equipment shall automatically resume when the externally induced magnetic field is removed.

3.10 Reliability requirements. Each equipment delivered against this specification shall be subjected to a burn-in of not less than 100 hours at 40°C of which at least the last 30 hours shall be failure free.

3.10.1 Reliability. The equipment reliability shall comply with the Type III requirements of MIL-T-28800 and as specified herein. The specified Mean Time Between Failure (MTBF) shall be 2,000 hours (See 6.2.3.1).

3.11 Maintainability requirements. The equipment shall comply with the Type III maintainability requirements of MIL-T-28800 and as specified herein (See 6.2.3.2).

3.11.1 Certification time. Certification time shall not exceed 0.5 hours, and it shall include all time required to check all inputs and outputs, on all range and function settings of the vertical (both channels shall be tested) and the horizontal section.

3.11.2 Maintenance provisions. The equipment shall comply with the Type III maintenance provisions requirements of MIL-T-28800 and herein.

3.11.2.1 Maintenance and calibration aids. Nonoperating accessories are circuit board extenders, special adapters, special tools, and patch cables required for maintenance and calibration of the equipment. Stowage of nonoperating accessories shall be as specified in 3.13.6.2. Operating accessories shall be as specified in 3.13.6.1.

3.11.2.2 Tilt bail handle assembly. The tilt bail handle assembly shall permit compliance with the enclosure attitude maintenance provision of MIL-T-28800. Removal of the tilt bail handle or handle assembly with common tools is permitted to comply with this requirement (See 6.2.2.1).

3.11.3 Indicator lights and light emitting diodes (LEDs). Unless approved by the procuring activity upon presentation of acceptable reliability data, indicator lights other than LEDs shall be accessible from the operator's side

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of the front panel (See 6.2.2.6). LEDs need not be accessible from the operator's side of the front panel.

3.11.4 Washability. The equipment shall comply with the washability requirements of MIL-T-28800 (See 6.2.2.7)

3.12 Interchangeability. The equipment shall comply with the Type III interchangeability requirements of MIL-T-28800.

3.13 Performance requirements. The equipment performance shall be over the 8 by 10 cm graticule area. Unless otherwise specified, all vernier controls shall be in the calibrated position and all performance requirements shall apply with or without the probes (See 4.5.2.1).

3.13.1 Warm-up. Unless otherwise specified herein, performance requirements apply to equipment after a 5 minute warm-up period at temperatures above 0°C and a 20 minute warm-up at temperatures below 0°C.

3.13.2 Cathode ray tube (CRT). A flat faced aluminized CRT having a useful scan of 10 divisions wide by at least 8 divisions high shall be provided. Unless otherwise specified in contract or order, the CRT shall utilize Type P31 phosphors (See 6.2.1.6). A division shall equal one cm.

3.13.2.1 Internal graticule. An internal graticule precision ruled in squares, 10 cm wide by at least 8 cm high shall be funrished. The horizontal and vertical centerlines shall be ruled in 0.2 cm minor divisions. The 10 and 90 percent points of a 5 cm or greater vertical display shall be etched across the horizontal display in readily discernible dots or marks.

3.13.2.2 Spot size. The spot shall be adjusted for optimum viewing at a point along the 8 by 10 cm graticule perimeter using equipment controls. Only the vertical and horizontal position controls may be adjusted to meet the following requirement: Moving the trace spot to any point on the perimeter of the 8 by 10 cm graticule shall not cause the greatest dimension of the trace spot to exceed 0.5 mm at any point on the perimeter of the 8 by 10 cm graticule. The trace spot when located at the center of the graticule shall be smaller than at any point on the perimeter by the 8 by 10 cm graticule.

3.13.2.3 Visual writing rate. The visual writing rate shall be such that a 10 nanosecond (NSEC) risetime pulse at a pulse repetition rate of 15 Hz shall be visible when tested in accordance with 4.5.3.

3.13.2.4 Z axis input (intensity modulation). A female BNC connector shall be provided on the equipment to permit intensity modulation over the DC to 15 MHz range. Intensity modulation shall be accomplished at optimum intensity with a Z axis input of from negative 5 Volts (to intensify) to positive 5 Volts (to blank). continuous operation maximum input without damage shall be ±20 Volts (DC + peak AC).

3.13.2.5 Bezel. A bezel shall be provided. The bezel shall be suitable for mounting a trace recording camera or viewing hood.

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3.13.2.6 CRT controls.

3.13.2.6.1 Focus. A front panel focus control shall be provided. The control and circuitry shall provide sufficient resolution for a clear sharp image at all degrees of intensity below the point of beam halo.

3.13.2.6.2 Beam finder. A front panel push button beam finder shall be provided for locating the electron beam position when it is not visible. When the beam finder is depressed the beam trace shall be confined to the face of the CRT. It shall be possible to center the beam with the horizontal and vertical position controls while the beam finder is depressed.

3.13.2.6.3 Astigmatism. An astigmatism control or circuitry shall be provided to control the roundness of the electron beam spot. The astigmatism control, if provided, shall be mounted on the front panel of Configuration A and C equipment.

3.13.2.6.4 Intensity. A front panel intensity control shall be provided. It shall provide for optimum intensity of the displayed signal and shall provide continuous range from extinguished beam to at least greater than the point of beam halo.

3.13.2.6.5 Trace rotation. A control of the recessed common screwdriver type shall be provided to rotate the trace about the horizontal axis. Rotation range shall be adequate to align the trace to the horizontal graticule lines. Sufficient offset shall be introduced to compensate for variations in the earth's magnetic field with at least ±2° overcompensation. The trace rotation control shall be mounted on the front panel of Configuration A and C equipment.

3.13.2.6.6 Graticule illumination. A front panel control shall be provided. It shall provide variable graticule illumination from zero to greater than optimum illumination.

3.13.2.7 Configuration A, Z axis input connector. The connector shall be located on the equipment front panel (See 3.3.2).

3.13.2.8 Configuration B, C, and D, Z axis input connector. The connector shall be located on the equipment front or rear panel (See 3.3.2).

3.13.3 Vertical input channels. Two channels shall be provided.

3.13.13.1 Bandwidth. The bandwidth of each vertical channel shall be from DC to at least 100 MHz with and without the probes attached (See 3.13.6.1.1). The frequency response referenced to a frequency of 50 kHz shall be constant at all deflection factors for all specified coupling modes. The frequency response shall approximate a Gaussian curve. Aberrations across the bandwidth shall be not greater than ±0.5 dB.

3.13.3.1.1 DC coupled bandwidth. The response shall be not more than 3 dB down at 100 MHz and shall be within ±0.25 dB from DC to 50 kHz. Below -15°C, the DC to 50 kHz bandwidth shall be within ±0.5 dB and the 50 kHz to 100 MHz bandwidth shall be not more than 3 dB down at 100 MHz.

3.13.3.1.2 AC coupled bandwidth. The frequency response curve shall be not more than 3 dB down at 10 Hz and 100 MHz. Below -15°C, the 10 Hz to 50 kHz bandwidth shall be within ±0.5 dB and the 50 kHz to 100 MHz bandwidth shall be not more than 3 dB down at 100 MHz.

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3.13.3.1.3 Frequency rolloff. The frequency response curve shall roll off smoothly (maximum negative slope of ±0.5 dB aberrations) between 100 MHz and 150 MHz for both AC and DC coupling. Below -15°C, frequency response roll off beyond 100 mHz shall have a negative slope with a maximum of ±1.0 dB aberrations.

3.13.3.1.4 Bandwidth measurements. Bandwidth measurements are specified with the equipment driven from a terminated 50 ohm source at the vertical input.

3.13.3.2 Transient response. The transient response aberrations of each vertical channel shall comply with the following using a step input pulse with a risetime not greater than 0.5 nsec (See 4.5.1.1).

3.13.3.2.1 Transient response aberrations. The transient response aberrations shall be within:

a. 2 percent p-p for a 6 cm step input between 15°C and 35°C.

b. 3 percent p-p for a 6 cm step input between -40°C and 15°C and between 35°C and 55°C.

3.13.3.3 Deflection factors. The minimum range for calibrated vertical channel deflection factors of each channel shall be from 5 millivolts (mV) per division or less to 5 Volts per division or more for both AC and DC coupling. Ranging shall be in 5-2-1 sequence.

3.13.3.3.1 Uncalibrated vertical vernier. The deflection factor shall be continuously variable between all ranges and extend the maximum deflection factor to at least 12 Volts per division.

3.13.3.3.2 Overload protection. The equipment shall be capable of withstanding at any vertical range setting, without damage and with the vertical channels AC or DC coupled up to:

a. ±250 Volts (DC + peak AC) to 20 kHz.

b. ±10 Volts (DC + peak AC) to 1 MHz.

c. ±5 Volts (DC + peak AC) to 100 MHz.

3.13.3.3.3 Vertical deflection factor accuracy. The vertical deflection factor DC accuracy of each vertical channel and each vertical deflection factor range setting shall be within:

a. 3 percent at any temperature between 0°C and 40°C.

b. 4 percent at any temperature between -40°C and 0°C and between 40°C and 55°C.

3.13.3.4 Input impedance. The input impedance at all sensitivity settings of each vertical channel shall be 1 megohm ±2 percent shunted by no more than 22 picofarads (pf). Below -15°C, the input impedance shall be 1 megohm ±5 percent shunted by no more than 22 pf.

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3.13.3.5 Vertical channel risetime. The rise time of each vertical input channel shall not exceed 35 nsec with and without the probes attached when measured between the 10 and 90 percent points of a step response that is 6 cm high (See 3.13.6.1.1).

3.13.3.6 Channel separation. The separation between the vertical input channels with both attenuators on the most sensitive range shall be not less than:

a. 100:1 to 10 MHz.

b. 50:1 over the 10 MHz to 20 MHz range.

c. 25:1 over the 20 MHz to 50 MHz range.

d. 15:1 over the 50 MHz to 100 MHz range.

3.13.3.7 Common mode rejection ratio (CMRR). The CMRR referenced to a 6 cm amplitude sine wave with both attenuators on the same calibrated range shall be not less than:

a. 25:1 to 10 MHz.

b. 10:1 over the 10 MHz to 100 MHz range

3.13.3.8 Direct current drift. The CRT trace drift of each vertical channel at the 5 mV per division setting shall not exceed:

a. 0.1 cm per hour between 0°C and 55°C.

b. 0.5 cm per hour between -40°C and 0°C.

3.13.3.9 Delay circuit. The vertical amplifier signals shall be delayed to allow the entire leading edge of an internally triggered 0.5 nsec or faster risetime, 1 MHz square wave of one cm amplitude at 5 mV per division to be visible.

3.13.3.10 Dynamic range. Transient response aberrations of each vertical channel shall not exceed those measured at center screen by more than 1.0 percent when:

a. The top of 2 cycles of a 6 cm displayed symmetrical square wave is positioned on the bottom graticule line.

b. The bottom of the same 6 cm displayed symmetrical square wave is positioned on the top graticule line.

3.13.3.11 X-Y display. The equipment shall have an X-Y display capability through the vertical input channels.

3.13.3.11.1 X-Y display bandwidth. The Y axis bandwidth shall be the same as Channel A (See 3.13.3.1). The X axis - 3 dB bandwidth shall be at least 4 MHz.

3.13.3.11.2 X-Y display phase difference. Phase difference in the X-Y display mode shall be not greater than 3° to at least 50 kHz increasing to no more than 4° at 100 kHz with both attenuators calibrated and in the 5 mV per division positions.

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3.13.3.11.3 X-Y display range. The X-Y display capability shall be over the full range of the vertical attenuators.

3.13.3.12 Vertical channel front panels controls. Unless otherwise specified herein, all of the following controls and indicators shall be located on the front panel (See 3.13.3.12.6).

3.13.3.12.1 Coupling. A coupling control for each vertical input channel shall be provided. The coupling controls shall have an AC, DC and a ground position. When in the ground position, the input signal circuit shall be opened and the input circuit to the input amplifier shall be grounded.

3.13.3.12.2 Attenuators. An attenuator for each vertical channel shall be provided (See 3.13.3.3).

3.13.3.12.3 Vertical position. A vertical position control, for each channel, shall be provided (See 3.13.3.10).

3.13.3.12.4 Display mode selector. The mode selector switch(s) shall provide the capabilities of a through e. This switch function may be combined with the requirements of 3.13.4.10.12.

a. Channel A only.

b. Channel B only.

c. Alternate. This mode shall alternately display the trace from Channel A on one sweep and the trace from Channel B on the next sweep for sweep speed settings of at least 0.5 millisecond (msec) per division to the maximum sweep speed.

d. Chopped. This mode shall display the channels alternating at one of the following modulation frequencies: 250 kHz, 400 kHz, or 1 MHz for sweep speed settings of at least 1 msec per division to the minimum sweep speed.

e. Algebraic. This mode shall display the instantaneous algebraic sum of Channel A and B.

3.13.3.12.5 Polarity inverter. A control shall be provided to invert the waveform being processed through at least one of the channels.

3.13.3.12.6 Direct current balance. A front panel or operator accessible DC balance control shall not be provided. The vertical channels shall be capable of retaining the DC balance within ±0.5 minor division over the attenuator range.

3.13.3.12.7 X-Y display mode. Control(s) shall be provided to permit operation of the vertical channels in an X-Y display mode (See 3.13.3.11).

3.13.3.12.8 Vertical channel uncalibrated vernier indicator. An indicator shall be provided for each vertical channel to indicate when the vertical channel uncalibrated vernier control is not in the calibrated position.

3.13.3.13 Vertical channel input connectors. Each vertical channel shall be provided with a front panel input connector (See 3.3.2).

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3.13.3.14 Bandwidth limiting. A control shall be provided to limit the vertical response at the -3 dB point of both channels to 20 ±5 MHz. This capability shall eliminate high frequency interference without affecting normal low frequency performance of the equipment.

3.13.4 Horizontal section.

3.13.4.1 Sweep trigger modes. The sweep trigger shall be selectable in any of the following trigger modes:

3.13.4.1.1 Normal. The sweep shall be selectably triggered by an internal vertical amplifier signal, external signal, or internal power line signal. A bright baseline shall be provided only in presence of trigger signal.

3.13.4.1.2 Automatic. A bright baseline shall be displayed at the maximum repetition rate for the range selected in the absence of input signals. Triggering shall be the same as in normal mode above 40 Hz.

3.13.4.1.3 Single (main time base only). The sweep shall occur once with the same triggering as normal. A capability to rearm the sweep and illuminate a reset lamp shall be provided. The sweep shall activate when the next trigger is applied, after rearming.

3.13.4.2 Retrace blanking. Retrace blanking shall be provided.

3.13.4.3 Time base (main). The time base shall provide calibrated sweep times in a 5-2-1 sequence from 0.05 microsecond (usec) per division or less to 0.5 second per division or more.

3.13.4.3.1 Time base accuracy. The calibrated sweep time shall be ±2 percent of the setting (below -15°C, the accuracy shall be within 4 percent of the setting).

3.13.4.3.2 Time base uncalibrated sweep vernier. The capability to provide uncalibrated sweep times calibrated steps and up to 1.5 seconds per division shall be provided.

3.13.4.3.3 Horizontal linearity. In the times (X) 1 position and with one time mark per division displayed and the first and eleventh marks exactly on the first and eleventh vertical graticule lines, the marks in between shall coincide with their respective graticule lines to within ±0.25 minor division.

3.13.4.4 Delayed time base. The delayed time base shall provide calibrated sweep times in a 5-2-1 sequence from 0.05 usec per division or less to 20 msec per division or more.

3.13.4.4.1 Delayed time base accuracy. The delayed time base accuracy shall be as specified in 3.13.4.3.1.

3.13.4.4.2 Delayed time base horizontal linearity. The delayed time base horizontal linearity shall be as specified in 3.13.4.3.3.

3.13.4.4.3 Delayed time base jitter. With the horizontal magnifier in the X1 or X10 position, jitter shall be not greater than 1 part in 20,000 (0.005 percent) of the main time per division setting.

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3.13.4.5 Trace intensification. The capability shall be provided to intensify that part of the main time base which is to expanded to full screen in the delayed time base mode.

3.13.4.6 Mixed sweep. A mixed sweep capability to combine the main and delayed sweep into one display shall be provided. This sweep shall be started by the main time base and completed by the faster delayed time base. Start of delayed time base portion of sweep shall be determined by delay time control.

3.13.4.7 Horizontal triggering.

3.13.4.7.1 Trigger level and slope (main and delayed time base).

a. Internal. The internal trigger level and slope shall permit triggering at any point on the positive or negative slope of a full screen displayed waveform.

b. External. The external level and slope shall permit continuously variable triggering from at least +1.0 Volt to -1.0 Volt on either slope of the trigger signal. A divide by 10 capability to extend triggering from at least +10.0 Volts to -10.0 Volts on either slope of the trigger signal shall also be provided.

3.13.4.7.2 Trigger source (main and delayed time base). The trigger source shall be selectable from the following:

a. Channel A. All displays shall be triggered by the Channel A signal.

b. Channel B. All displays shall be triggered by the Channel B signal.

c. Composite. All displays shall be triggered by the combined waveforms from Channels A and B.

3.13.4.7.3 Trigger sensitivity (main and delayed time base).

a. Internal. A stable sweep shall be obtained from a sine wave having at least 0.3 cm p-p vertical deflection from DC to at least 25 MHz, increasing to 1.0 cm p-p vertical deflection at 100 MHz.

b. External. A stable sweep shall be obtained from a sine wave having at least 50 mV p-p vertical deflection from DC to 25 MHz, increasing to 150 mV p-p vertical deflection at 100 MHz.

3.13.4.7.4 External trigger input (main and delayed time base).

a. Input impedance. The external trigger input impedance shall be 1 megohm ±5 percent paralleled by not more than 35 pf.

b. Maximum input. The external trigger input shall withstand, without damage, at least ±100 Volts (DC + peak AC).

c. Coupling. The selectable AC and DC coupling capability shall also be provided.

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3.13.4.7.5 Trigger jitter (main and delayed time base). Trigger jitter shall not exceed 0.5 nsec at 100 MHz and 5 nsec per division with the X10 magnifier on.

3.13.4.8 Sweep expansion (horizontal magnifier) (main and delayed time base). A X10 sweep expansion capability shall be provided.

3.13.4.8.1 Sweep expansion accuracy.

a. X10 accuracy shall be within ±5 percent of the setting including time base accuracy.

b. The center of the trace shall not shift more than 2.0 cm when the horizontal magnifier is changed from X1 to X10.

3.13.4.9 Horizontal position drift (main and delayed time base). The CRT horizontal position drift with the horizontal magnifier in the X1 position and at any temperature shall not exceed.

a. 0.1 cm per hour between 0°C and 55°C.

b. 0.5 cm per hour between -40°C and 0°C.

3.13.4.10 Horizontal channel front panel controls. Unless otherwise specified herein, all of the following controls and indicators shall be located on the front panel.

3.13.4.10.1 Time base mode selector. The following time base modes shall be provided:

a. Normal sweep. This mode shall permit operation in the normal undelayed manner.

b. Delayed sweep. This mode shall permit the generation of a sweep after a defined interval of delay following the trigger pulse.

c. Mixed sweep. This mode shall permit the combining of the main and delayed sweep into one display.

3.13.4.10.2 Main time base selector. A control shall be provided to select the calibrated sweep times of 3.13.4.3.

3.13.4.10.3 Main time base uncalibrated sweep vernier lamp. An indicator shall be provided to indicate when the main time base nernier is not in the calibrated position.

3.13.4.10.4 Delayed time base selector. A control shall be provided to select the calibrated sweep times of 3.13.4.4.

3.13.4.10.5 Variable sweep delay time. A 200 nsec or less to 0.5 seconds or more delay time control shall be provided by a 10 turn control with 10 major and 100 minor divisions.

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3.13.4.10.5.1 Differential time measurement accuracy. The differential time measurement accuracy shall be within 1.0 percent of full scale for measurements over 2 or more major dial divisions excluding delay time dial position settings between 0.0 and 0.1 and excluding delaying time per division switch positions of 0.05 usec per division over the +15°C to +35°C temperature range. Over the 0°C to +15°C and 35°C to 55°C temperature ranges an additional 1.0 percent of degradation is allowed. Below 0°C temperature and referenced to the +15°C to 35°C temperature range, an additional 3.0 percent degradation is allowed.

3.13.4.10.6 Trigger coupling (main and delayed time base). Control shall be provided for the following types of trigger couplings.

a. Direct current (DC). This type of coupling shall be provided for coupling signals from DC to 100 MHz.

b. Alternating current (AC). This type of coupling shall incorporate a high pass filter which attentuates signals less than 30 Hz.

c. Low frequency rejection. This type of coupling shall incorporate a high pass filter which attenuates signals less than 15 kHz (main time base only).

d. High frequency rejection. This type of coupling shall incorporate a low pass filter which attenuates signals greater than 50 kHz (main time base only).

3.13.4.10.7 Trigger level (main and delayed time base). A continuously variable control(s) shall be provided to permit triggering as specified in 3.13.4.7.

3.13.4.10.8 Trigger slope (main and delayed time base). A control shall be provided to permit triggering as specified in 3.13.4.7.

3.13.4.10.9 Sweep trigger mode (main and delayed time base). Sweep trigger mode control(s) shall be provided to comply with the requirement of 3.13.4.1.

3.13.4.10.10 Trigger source (main and delayed time base). A control(s) shall be provided to select a trigger, automatic and selectable, from internal vertical amplifiers, external signals or internal power line sources. The internal line trigger source need not be provided for the delayed time base.

3.13.4.10.11 Horizontal position. A horizontal position control shall be provided to move the left hand end of the trace to the right past the center graticule line in the X1 and X10 horizontal magnifier position and the right hand end of the trace to the left past the center graticule line in the X1 and X10 horizontal magnifier position.

3.13.4.10.12 Channel trigger source (main and delayed time base). A trigger source selector shall be provided to permit the selection of triggering from channel A only, channel B only, and composite. This switch may be combined with the requirements of 3.13.3.12.4.

3.13.4.10.13 Divide by 10 (main and delayed time base). Controls shall be provided to divide by 10 for the external trigger input.

3.13.4.10.14 Sweep expansion (horizontal magnifier). A X10 sweep expansion control shall be provided.

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3.13.4.10.15 Trigger hold-off. A trigger hold-off control shall be provided to permit the sweep period to be continuously variable to at least 3.0 times the time per division of each horizontal range from 20 msec per cm through 50 msec per cm.

3.13.4.10.16 Reset push button and lamp. A reset push button shall be provided to arm the sweep while in single sweep operation. The reset lamp shall indicate when the sweep is armed and in single sweep operation.

3.13.4.10.17 Trace intensification. A control shall be provided to select the trace intensification mode specified in 3.13.4.5.

3.13.4.10.18 Trigger view. A control shall provide a vertically calibrated display of the signal being used for the main time base triggering and point of triggering. The trigger shall be displayed with the same sweep selected by the operator for viewing the vertical signal. The deflection factor of the trigger view amplifier shall be 50 or 100 mV per cm. The point of triggering shall be identified by the crossover of the center horizontal graticule line by the trigger signal.

3.13.4.11 Horizontal channel connectors. The following connectors shall be provided in accordance with 3.3.2.

3.13.4.11.1 External trigger input connector. Front panel external trigger input connectors shall be provided.

3.13.4.11.2 Horizontal output connectors. The output connectors of the Configuration A and C equipment shall be mounted on the front panel. The output connectors of the Configuration B and D equipment shall be mounted on the front or rear panel. The following horizontal outputs shall be provided.

a. Main gate. The gate shall be coincident with the main sweep wave form. The gate shall be at least positive 0.5 Volts peak or negative 0.5 Volts into 1 megohm paralleled by up to 25 pf.

b. Delayed gate. The gate shall be coincident with the delayed sweep waveform. The gate shall be at least positive 1.0 Volts peak or negative 1.0 Volts into 1 megohm paralleled by up to 25 pf.

3.13.5 Calibrator. A calibrator signal shall be provided through a connector mounted on the front panel. The connector shall be compatible with one of the probe tips specified in (3.13.6.1.1.5). The calibrator signal shall be 1, 2.5 or 5 kHz ±10 percent square wave having an amplitude of 1.0 Volts p-p and the characteristics of 3.13.5.1. The calibrator voltage shall be regulated to within ±1 percent when loaded by 1 megohm paralleled by up to 25 pf.

3.13.5.1 Calibrator characteristics.

a. Risetime: Not greater than 1.0 usec.

b. Symmetry: Within ±25 percent.

c. The calibrator square wave shall be flat-topped with no noticeable overshoot or leading edge rounding.

d. The calibrator shall be short circuit proof.

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3.13.6 Accessories. The following accessories shall be provided with each equipment.

3.13.6.1 Operating accessories. Configuration A and B operating accessories shall be stowed as specified in MIL-T-28800 for applicable enclosure style. configuration C and D operating accessories shall be stowed in an external unattached accessories pouch which shall be furnished with the Configuration C and D oscilloscope.

3.13.6.1.1 Probes.

3.13.6.1.1.1 Probe compensation. Adjustable probe compensation consistent with the vertical channel input capacitance shall be provided in each probe as applicable.

3.13.6.1.1.2 Probe cable characteristics. Each probe assembly shall not be less than 1 m (39.4 inches) long, of appropriate characteristics and have maximum flexibility and longevity. A detachable ground lead approximately 0.15 m (6 inches) in length, terminated with an alligator clip, shall be provided.

3.13.6.1.1.3 10:1 Voltage divider probes. Two 10:1 voltage divider probes shall be provided for use at frequencies up to at least 100 MHz.

3.13.6.1.1.4 1:1 Voltage divider probe. One 1:1 voltage probe shall be provided for use at frequencies up to at least 25 MHz.

3.13.6.1.1.5 Probe tips. Each probe shall be provided with the following tips:

a. Pincer jaw (or retractable hook tip)

b. Banana tip.

c. Spring clip.

d. Hook tip.

e. Pin tip.

3.13.6.1.2 Adapters.

a. One Type UG-1035 ( ) /U or UG-1441/U

b. One Tee connector: Type UG-274C/U

3.13.6.1.3 Viewing hood. A viewing hood capable of being attached to the bezel shall be supplied only when specified in contract or order (See 6.2.9).

3.13.6.1.4 Light filters. A blue light filter shall be supplied to provide contrast and protect the CRT display surface.

3.13.6.1.5 Operating instructions. When required in the contract or order, each instrument shall be supplied with operating instructions. For Configurations A and B, these instructions shall be stored within the front panel cover (See 6.2.2.5).

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3.13.6.2 Nonoperating accessories. Unless otherwise specified in the contract or order, Configuration A, B, and C nonoperating accessories shall be stowed within the equipment case. For Configuration D, or when specificed in the contract or order, nonoperating accessories shall be stowed in a reusable pouch or container which shall be marked "Maintenance Kit for use with (equipment nomenclature or model)." The equipment nomenclature shall be marked thereon, if assigned (See 6.2.1.5, 6.2.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Bid Sample testing will be performed by the Government.

4.1.1 Quality assurance terms and definitions. Quality assurance terms used herein shall be in accordance with MIL-T-28800.

4.2 Classification of inspections. The inspections required herein are classified as in (a) through (c).

a. First Article inspection (See 4.2.1)

b. Bid Sample inspection (See 4.2.2)

c. Quality conformance inspection (See 4.2.3)

4.2.1 First Article inspection groups. First Article examinations and tests are classified into groups in accordance with Table I and (a) through (f) (See 4.3 and 4.3.1).

a. First Article inspection, Group A (See 4.3.1.1)

b. First Article tests, Group B (See 4.3.1.2)

c. First Article tests, Group C (See 4.3.1.3)

d. First Article tests, Group D (See 4.3.1.4)

e. First Article tests, Group E (See 4.3.1.5)

f. First Article tests, Group F (See 4.3.1.6)

4.2.2 Bid Sample inspection groups. Bid Sample examinations and tests are classified into groups in accordance with Table I and (a) through (f) (See 4.3. and 4.3.2).

a. Bid Sample inspection, Group A (See 4.3.2.1)

b. Bid Sample tests, Group B (See 4.3.2.2)

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c. Bid Sample tests, Group C (See 4.3.2.3)

d. Bid Sample tests, Group D (See 4.3.2.4)

e. Bid Sample tests, Group E (See 4.3.2.5)

f. Bid Sample tests, Group F (See 4.3.2.6)

4.2.3 Quality conformance inspection groups. Quality conformance examinations and tests are classified into groups in accordance with Table I and (a) through (f) (See 4.4)

a. Production inspection, Group A (See 4.4.1)

b. Production control tests, Group B (See 4.4.2)

c. Environmental and electrical power tests, Group C (See 4.4.3)

d. Environmental tests, Group D (See 4.4.4)

e. Other tests, Group E (See 4.4.5)

f. Reliability tests, Group F (See 4.4.6)

TABLE I. Examination and test groups.

Description	Requirement	Inspection Method: First Article	Bid Sample	Quality Conformance
Group A				
Preoperational Inspection	3	4.5	4.5	4.5
Level A Performance	3.13	4.5	4.5	4.5
Leakage Current	3.2	4.5	4.5	4.5
Visual Writing Rate	3.13.2.3	4.5.3	4.5.3	4.5.3
Group B				
Level B Performance	3.13	4.5	4.5	4.5
Group C				
Temperature and Humidity				
Humidity	3.9	4.5	4.5	4.5
Low Temperature	3.9	4.5.4	4.5.4	4.5.4
High Temperature	3.9	4.5	4.5	4.5
Altitude	3.9	4.5	4.5	4.5
Vibration	3.9	4.5	4.5	4.5
Shock, Mechanical				
Shock, Pulse	3.9	4.5	4.5	4.5
Transit Drop, Configuration A&B	3.9	4.5.5	4.5.5	4.5.5
Bench Handling	3.9	4.5	4.5	4.5
Water Resistance				
Watertight, Configuration A	3.9	4.5	4.5	4.5
Splashproof, Configuration A	3.9	4.5	4.5	4.5
Dripproof, Configuration B	3.9	4.5	4.5	4.5

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TABLE I. Examination and test groups. - Continued

Description	Requirement	Inspection Method: First Article	Bid Sample	Quality Conformance
Electrical Power				
Input Power Consumption	3.5.1	4.5	4.5	4.5
Voltage and Frequency Variation				
Configuration A	3.5.2.1	4.5	4.5	4.5
Configuration B,C, and D	3.5.3.1	4.5	4.5	4.5
Voltage Transient				
Configuration A	3.5.2.1	4.5	4.5	4.5
Configuration B,C, and D	3.5.3.1.1	4.5	4.5	4.5
Frequency Transient				
Configuration A	3.5.2.1	4.5	4.5	4.5
Configuration B,C, and D	3.5.3.1.1	4.5	4.5	4.5
Verification, Group C	3.13	4.5	4.5	4.5
Group D				
Fungus Resistance	3.9	4.5.6	4.5.6	4.5.6
Salt Atmosphere Exposure				
Structural Parts				
Configuration A	3.9.1.1	4.5	4.5	4.5
Configuration B	3.9.2.1	4.5	4.5	4.5
Configuration C	3.9.3.1	4.5	4.5	4.5
Enclosure, Configuration A	3.9.1.1	4.5	4.5	4.5
Explosive Atmosphere, Configuration A	3.9.1	4.5.7	4.5.7	4.5.7
Sand and Dust, configuration A	3.9.1	4.5	4.5	4.5
Electromagnetic Compatibility				
Electromagnetic Interference	3.9.5.1	4.5.8.1	4.5.8.1	4.5.8.1
Magnetic Environment	3.9.5.2	4.5.8.2	4.5.8.2	4.5.8.2
Verification, Group D, Configuration A	3.13	4.5	4.5	4.5
Group E				
Maintainability	3.11	4.5	4.5	4.5
Washability	3.11.4	4.5	4.5	4.5
Acoustic Noise	3.9	4.5	4.5	4.5
Dimensions	3.6	4.5	4.5	4.5
Weight	3.6	4.5	4.5	4.5
Mechanical Stability	3.6	4.5	4.5	4.5
X-Radiation (Radiant Energy)	3.2	4.5	4.5	4.5
Front Panel Marking	3.8	4.5	4.5	4.5
Oscilloscope and Battery Pack Compatibility, Configuration A	3.5.2.1	4.5.9	4.5.9	4.5.9
Group F				
Reliability	3.10	4.5	4.5	4.5

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4.3 First article or bid sample inspection. Unless otherwise specified, test plans and test reports; test methods, procedures, conditions, and control settings shall be as specified in 4.5 and MIL-T-28800 with the equipment oriented for horizontal viewing.

4.3.1 First article inspection. When specified in solicitation, the first article inspection shall be performed on seven equipments and a sample of the front panel (See 6.2.2.8 and 6.2.2.9).

4.3.1.1 First article inspection, Group A. All seven equipments that have satisfactorily passed the Group A inspection of Table I.

4.3.1.2 First article tests, Group B. The seven equipments that have satisfactorily passed the Group A inspection shall be subjected to the Group B Tests of Table I.

4.3.1.3 First article tests, Group C. Two equipments that have satisfactorily passed the Group B tests shall be subjected to the Group C tests of Table I. These shall not be the two equipments selected for Group D testing.

4.3.1.4 First article tests, Group D. Two equipments that have satisfactorily passed the Group B tests shall be subjected to the Group D tests of Table I. These shall not be the two equipments selected for Group C testing.

4.3.1.5 First article tests, Group E. Equipments that have satisfactorily passed the Group B tests (See 4.3.1.2) shall be subjected to the Group E tests of Table I. The number of equipments to be tested shall be as in a through f.

a. Maintainability. Two equipments shall be tested. The washability test shall be performed on equipments prior to the maintainability demonstration.

b. Acoustical noise test. Two equipments shall be tested if forced air cooling is used.

c. Dimensions and weight tests. Two equipments shall be tested.

d. Front panel marking test. One sample of the front panel marking process shall be tested.

e. X-radiation test. Two equipments shall be tested if operating potentials of 15,000 Volts or more are used or if the failure of any one component or nonrepairable subassembly can result in any voltage increasing to 15,000 Volts or more.

f. Oscilloscope and battery-pack compatibility test. Two equipments shall be tested.

g. Mechanical stability. Two equipments shall be tested.

4.3.1.6 First article tests, Group F. Five equipments that have satisfactorily passed the Group B tests shall be subjected to the Group F tests of Table I.

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4.3.2 Bid sample inspection. When required in solicitation, Bid Samples shall be submitted to the procuring activity for Bid Sample inspection. Unless otherwise required in the solicitation, the Bid Sample shall consist of two equipments, including accessories, instructions manuals and an unperforated 100 mm (4 inch) by 100 mm (4 inch) sample of the front panel marking process (all representative of production equipment) and clear and concise rationale showing how the reliability and maintainability characteristics of the equipment comply with the requirements of the solicitation (See 6.2.1.9 and 6.3).

4.3.2.1. Bid sample inspection, Group A. The two equipments will be subjected to the Group A inspection of Table I.

4.3.2.2 Bid sample tests, Group B. The two equipments that have satisfactorily passed the Group A inspection will be subjected to the Group B tests of Table I.

4.3.2.3 Bid sample tests, Group C. One equipment that has satisfactorily passed the Group B tests will be subjected to the Group D tests of Table I (See 4.3.2.7).

4.3.2.4 Bid sample tests, Group D. One equipment that has satisfactorily passed the Group B tests will be subjected to the Group D tests of Table I (See 4.3.2.7).

4.3.2.5 Bid sample tests, Group E. Equipments that have satisfactorily passed the Group B tests will be subjected to the Group E tests of Table I. The number of equipments to be tested will be as in a through e.

a. Maintainability. No equipment will be tested.

b. Acoustical noise test. One equipment will be tested if forced air cooling is used.

c. Dimensions and weight tests. One equipment will be tested.

d. Front panel marking test. One sample of the front panel will be tested.

e. X-radiation test. One equipment will be tested if operating potentials of 15,000 Volts or more are used or if the failure of any one component or nonrepairable subassembly can result in any voltage increasing to 15,000 Volts or more.

f. Mechanical stability. One equipment will be tested.

4.3.2.6 Bid Sample tests, Group F. The equipment will not be subjected to reliability testing.

4.3.2.7 In the event a Bid Sample becomes unacceptable during the Group C or Group D testing, the remaining Bid Sample(s) will also be subjected to the Group C or Group D tests which were not satisfactorily completed by the unacceptable Bid Sample(s).

4.4 Quality conformance inspection. Unless otherwise required herein, the test methods, procedures, and conditions used for quality conformance inspection shall be as specified in 4.5 and MIL-T-28800. (See 6.2.2.10 and 6.2.2.11)

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4.4.1 Production inspection. Each equipment shall be subjected to the Group A inspection of Tables I and II to ensure qualitatively the proper functioning of the equipment, including all operating controls, and conformance with safety requirements.

4.4.2 Production control tests. The Group B tests of Tables I and III shall be performed on equipment that have passed Group A inspection. (See 4.4.1)

4.4.2.1 Production control sampling plan. From each lot a random sample of equipment shall be selected and subjected to the Group B tests. Minimum sampling shall be in accordance with MIL-T-28800.

4.4.3 Environmental and electrical power tests. The Group C tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.3.1 Environmental and electrical power sampling plan. From each lot a random sample of equipment shall be selected and subjected to the Group C tests. The sampling shall be in accordance with MIL-T-28800.

4.4.4 Environmental tests. The Group D tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.4.1 Environmental tests sampling plan. From each lot a random sample of equipment shall be selected and subjected to the Group D tests. The sampling shall be in accordance with MIL-T-28800.

4.4.5 Other tests. The Group E tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2.

4.4.5.1 Other tests sampling plan. From the first production lot, a random sample shall be selected and subjected to the Group E tests. The sampling shall be in accordance with MIl-T-28800.

4.4.6 Reliability tests. The Group F tests of Table I shall be performed on equipment which have passed the required tests in accordance with 4.4.2. The Group F tests shall be conducted in accordance with the Type III reliability tests of MIL-T-28800. Burn in for each reliability sample unit shall be a minimum of 100 hours with the last 30 hours failure free. If a failure occurs within the 30 hour failure free period, burn in on that unit will stop, repair will be made, and an additional 30 hours burn in will be added to the total accrued testing time for that unit. Additional failures shall result in additional burn in extensions until the final 30 hours of burn in are failure free.

4.4.6.1 Reliability demonstration phase. Unless reliability has been demonstrated in accordance with 4.3.1.6, the demonstration phase of the Type III reliability test shall be performed on a random sample of ten equipment selected from the first production lot. If reliability has been demonstrated in accordance with 4.3.1.6, the first production lot need not undergo the demonstration phase of the Type III reliability test.

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4.4.6.2 Reliability sampling phase. If the reliability has been demonstrated in accordance with 4.3.1.6, the sample phase of the Type III reliability tests shall be performed on a random sample of 10 equipment selected from each production lot. If reliability has been demonstrated on the first production lot in accordance with 4.4.6.1, subsequent production lots shall undergo the sampling phase of the Type III reliability test.

4.5 Examination and test methods. The examination and test methods, procedures, conditions and fialure criteria of MIL-T-28800 shall be applied except as modified herein (See Tables I, II, and III). When the equipment is identified by configuration it shall be tested in accordance with the applicable type, class, and style requirements of MIL-T-28800 as specified in 3.1.1 through 3.1.4.

The equipment performance during the operating portions of Groups C through F testing shall be verified in accordance with (a) and (b).

a. Satisfactory operation check. The satisfactory operation check shall be as follows:

(1) Display the calibrator output on Channel A.

(2) Display a similar waveform on Channel B (from a source other than the equipment calibrator).

(3) Set Channel A and B vertical attenuators to the same setting.

(4) Set the equipment controls for optimum display.

(5) Set the vertical mode selector for alternate display.

(6) Set to internal AC triggering.

(7) Observe that proper operation of the equipment is manifested.

b. Satisfactory operation test. When specified in the contract or order, the contractor shall prepare a satisfactory operation test procedure as part of the detailed test procedure. The satisfactory operation test procedure shall comprise such tests as will reveal degradation in function and performance beyond that permitted by the specification and shall include as a minimum those indicated in Table II (Satisfactory Operation Test) (See 6.2.2.16).

4.5.1 Test equipment. When specified in the contract or order, the contractor shall prepare a list of test equipment to be used for inspections as part of the detailed test procedure (See 6.2.2.17). Test equipment approval for deviation from the prepared list shall be in accordance with MIL-T-28800 and as provided for in the contract or order.

4.5.1.1 Test equipment accuracy and calibration. Test equipment to be used for testing the equipment shall have an accuracy of at least five times better than the equipment parameter under test wherever possible. Calibration of the test equipment shall be in accordance with MIL-T-28800 and as provided for in the contract or order. (See 6.2.3.3)

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TABLE II. Level A performance test parameters.

Parameter	Requirement	Satisfactory Operation Test
Probe Compatibility	*3.13	No
Warm-Up	3.13.1	Yes
Spot Size	3.13.2.2	No
Focus	3.13.2.6.1	No
Beam Finder	3.13.2.6.2	No
Astigmatism	3.13.2.6.3	No
Intensity	3.13.2.6.4	No
Bandwidth	*3.13.3.1	No
DC Coupled Bandwidth	3.13.3.1.1	No
AC Coupled Bandwidth	*3.13.3.1.2	*Yes
Transient Response	3.13.3.2	Yes
Deflection Factors	*3.13.3.3	No
Vertical Deflection Factor Accuracy	*3.13.3.3.3	No
Vertical Channel Risetime	3.13.3.5	Yes
Delay Circuit	3.13.3.9	No
Dynamic Range	3.13.3.10	Yes
X-Y Display Phase Difference	3.13.3.11.2	No
Coupling	3.13.3.12.1	No
Vertical Position	3.13.3.12.3	No
Polarity Inverter	3.13.3.12.5	No
Vertical Uncalibrated Indicator	3.13.3.12.8	No
Sweep Trigger, Normal	3.13.4.1.1	No
Sweep Trigger, Automatic	3.13.4.1.2	No
Retrace Blanking	3.13.4.2	No
Time Base	*3.13.4.3	No
Time Base Accuracy	*3.13.4.3.1	No
Time Base Horizontal Linearity	*3.13.4.3.3	Yes
Trigger Level and Slope	3.13.4.7.1	No
Trigger Source	3.13.4.7.2	No
Trigger Sensitivity	3.13.4.7.3	Yes
Trigger Jitter	3.13.4.7.5	No
Uncalibrated Sweep Vernier Indicator	3.13.4.10.3	No
Horizontal Position	3.13.4.10.11	Yes
Channel Trigger Source	3.13.4.10.12	No
Calibrator	*3.13.5	Yes
Graticule Illumination	3.13.2.6.6	Yes
Bandwidth Limiting	3.13.3.14	No
Sweep Trigger, Single	3.13.4.1.3	No
Mixed Sweep	3.13.4.6	Yes
Delayed Time Base	3.13.4.4	Yes
Delayed Time Base Accuracy	3.13.4.4.1	No
Delayed Time Base Linearity	3.13.4.4.2	No
Delayed Time Base Jitter	3.13.4.4.3	No
Trace Intensification	3.13.4.5	Yes
Trigger View	3.13.4.10.18	Yes

* Abbreviated Testing

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TABLE III. Level B performance test parameters.

Parameter	Requirement
Warm-Up	3.13.1
Z Axis Input	3.13.2.4
Trace Rotation	3.13.2.6.5
Bandwidth	3.13.3.1
DC Coupled Bandwidth	3.13.3.1.1
AC Coupled Bandwidth	3.13.3.1.2
Frequency Rolloff	3.13.3.1.3
Deflection Factors	3.13.3.3
Uncalibrated Vertical Vernier	3.13.3.3.1
Overload protection	3.13.3.3.2
Vertical Deflection Factor Accuracy	3.13.3.3.3
Input Impedence	3.13.3.4
Channel Separation	3.13.3.6
Common Mode Rejection Ratio (CMRR)	3.13.3.7
Direct Current Drift	3.13.3.8
X-Y- Display	3.13.3.11
X-Y Display Bandwidth	3.13.3.11.1
X-Y Display Sensitivity	3.13.3.11.3
Display Mode Selector	3.13.3.12.4
Direct Current Balance	3.13.3.12.6
Time Base	3.13.4.3
Time Base Accuracy	3.13.4.3.1
Uncalibrated Sweep Vernier	3.13.4.3.2
Time Base Horizontal Linearity	3.13.4.3.3
External Trigger Input	3.13.4.7.4
Sweep Expansion Accuracy	3.13.4.8.1
Horizontal Position Drift	3.13.4.9
Horizontal Position	3.13.4.10.11
Divide by 10	3.13.4.10.13
Calibrator	3.13.5
Mixed Sweep	3.13.4.6
Delayed Time Base	3.13.4.4
Delayed Time Base Linearity	3.13.4.4.2
Variable Sweep Delay Time Differential Time Measurement	3.13.4.10.5.1
Trigger Coupling	3.13.4.10.6
Trigger Holdoff	3.13.4.10.15
Reset Pushbutton and Lamp	3.13.4.10.16
Trigger View	3.13.4.10.18
Horizontal Outputs	3.13.4.11.2

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4.5.2 Conditions and control settings.

4.5.2.1 Test conditions. Unless otherwise required in the detailed test or herein, the test conditions shall be in accordance with MIL-T-28800 except that all performance requirements shall apply with or without the probes. The equipment vertical channels with probes attached shall be tested to demonstrate compliance with the requirements of 3.13.3.1, 3.13.3.5, and 3.13.5 at the probe tip. Abbreviated testing may be used, however, each probe shall be tested. Unless otherwise specified, the equipment shall be tested as oriented for horizontal viewing, normal operating positions as in (a) and (b).

a. Configurations A and B equipment normal operating positions provide for horizontal viewing of the CRT face for use on a bench and vertical viewing of the CRT face when the equipment is on a floor or deck.

b. Configuration C and D normal operating position provides for horizontal viewing such as when the equipment is rack-mounted (Configuration C) or located on a bench (Configuration D).

4.5.2.2 Control settings. The control settings shall be in accordance with the approved detailed test procedure (See 6.2.2.15) except all vernier controls shall be in the calibrated position.

4.5.3 Visual writing rate test. The visual writing rate test shall be performed with a 1.0 Volt, 10 ±1 nsec risetime pulse applied to the Channel A vertical input connector from a 50 ohm terminated source at a rate of 15 ±1 pulses per second. The controls of the equipment under test shall be adjusted to display the risetime portion of the pulse over the center 5 cm horizontal segment of the graticule and with the 10 and 90 percent points of the pulse located at the intersection of the etched 10 and 90 percent points of the graticule. The pulse characteristics including any ringing, preshoot or overshoot shall be visible and identifiable. A viewing hood may be used in the performance of this test.

4.5.4 Low temperature test. The low temperature test shall be performed in accordance with MIL-T-28800 except that an additional satisfactory operation test shall be performed at -14°C.

4.5.5 Transit drop test. The transit drop test shall be performed in accordance with the transit drop test of MIL-T-28800 for Configurations A and B, except the Configuration B drop height shall be 8 inches.

4.5.6 Fungus inert materials test. Samples of all parts, materials and finishes not known to be inherently fungus inert shall be subjected to the fungus inert materials test of MIL-T-28800 (See 6.2.2.18).

4.5.7 Explosive atmosphere test. Configuration A equipment shall be subjected to the explosive atmosphere test in accordance with MIL-T-28800. The vertical attenuators shall be switched during the explosive atmosphere test with maximum input voltage applied to the vertical input channel.

4.5.8 Electromagnetic compatibility test.

4.5.8.1 Electromagnetic interference (EMI) tests. The EMI tests shall be performed in accordance with MIL-T-28800 and as in (a) through (i).

a. CE 01: Narrow band requirement from 10 kHz to 20 kHz only.

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b. CE 03: Full limits

c. CS 01: Full limits

d. CS 02: Full limits

e. CS 06: Full limits

f. RE 01: Relaxed 10 dB at the fundamental, third harmonic, and fifth harmonic of the power source frequency.

g. RE 02: Full limits to 10 times the highest used or intentionally generated frequency or 1 GHz, whichever is greater.

h. RS 01: Full limits

i. RS 03: Full limits to 1 GHz.

4.5.8.2 Magnetic environment test. The magnetic environment test shall be performed in accordance with MIL-T-28800. Failure criteria shall be as specified herein (See 3.9.5.2).

4.5.9 Oscilloscope and battery-pack compatibility test. The oscilloscope and battery pack compatibility test shall be performed over the -20°C to +55°C temperature range in accordance with (a) through (e).

a. Place the equipment in a controled environment at -20°C and allow it to stabilize until the rate of change is 2°C per hour or less.

b. Connect the equipment to the battery-pack or other DC power source approved by the procuring activity (See 6.2.2.19).

c. Perform the satisfactory operation check (see 4.5) for a minimum of 5 minutes.

d. Record the power consumption and the results of the satisfactory operation check (See 6.2.2.20).

e. Repeat the test in 10 degrees C steps from -20°C to +50°C and at 55°C. The rate of change between steps shall be 20°C per hour. The relative humidity need not be controlled.

4.6 Optional quality assurance provisions. Upon approval by the procuring activity, a Department of Defense approved supplier's quality assurance system may be substituted for any of the tests in Section 4 (See 6.2.2.21).

5. PACKAGING

5.1 Preparation for delivery shall be in accordance with MIL-T-28800 and as specified by the procuring activity (See 6.2.1.11).

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6. NOTES

6.1 Intended use. These equipments are intended for use as general purpose oscilloscopes for testing and other uses with electronic and electrical equipment. Configuration A equipments are for use in uncontrolled environments. Configuration B, C, and D equipments are for use in laboratory, shop, repair, and depot activities.

6.2 Ordering data.

6.2.1 Procurement requirements listing. The procurement documents should specify:

6.2.1.1 Title, number, and date of this specification.

6.2.1.2 Configuration desired (See 3.1).

6.2.1.3 Finishes of Configuration A if other than forest green (See 3.3.3).

6.2.1.4 Power cable for 230 VAC operation (See 3.5.3.3., 3.5.3.2).

6.2.1.5 Nonoperating accessories (See 3.13.6.2).

6.2.1.6 Phosphor type if other than P31 (See 3.13.2).

6.2.1.7 Viewing hood type, if required (See 3.13.6.1.3).

6.2.1.8 First articles, if applicable (See 4.2.1).

6.2.1.9 Bid samples, if applicable (See 4.2.2., 4.3.2).

6.2.1.10 Reliability and maintainability requirements (to be supplied elsewhere in procurement document) (See 4.3.2).

6.2.1.11 Levels of packing and packaging other than specified (See 5.1).

6.2.2 Contract data requirements listing.

6.2.2.1 Tilt bail handle approval, Configuration B and D (See 3.7.3, 3.7.5, 3.11.2.2).

6.2.2.2 Nomenclature assignment (See 3.8, 3.13.6.2).

6.2.2.3 National stock number (See 3.8).

6.2.2.4 Engineering design change, approval and marking (See 3.8.1.3).

6.2.2.5 Operating instructions (See 3.13.6.1.5).

6.2.2.6 Indicator light reliability data (See 3.11.3).

6.2.2.7 Procedure for water-washing oscilloscopes (See 3.11.4).

6.2.2.8 First article test plans (See 4.3.1).

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6.2.2.9 First article test reports (See 4.3.1).

6.2.2.10 Quality assurance test palns (See 4.4).

6.2.2.11 Quality assurance test reports (See 4.4).

6.2.2.12 Pretest performance records (See 4.5).

6.2.2.13 Maintainability test records (See 4.5).

6.2.2.14 Reliability test records (See 4.5).

6.2.2.15 Detailed test procedure (See 4.5, 4.5.2.2).

6.2.2.16 Satisfactory operation test procedure (See 4.5).

6.2.2.17 Test equipment list (See 4.5.1).

6.2.2.18 Fungus inert materials certification (See 4.5.6).

6.2.2.19 DC source approval, Configuration A (See 4.5.9).

6.2.2.20 Battery-pack compatibility, Configuration A (See 4.5.9).

6.2.2.21 Optional quality assurance provisions (See 4.6).

6.2.3 Acquisition management systems. The following acquisition management systems are referenced herein.

6.2.3.1 Reliability (See 3.10.1).

6.2.3.2 Maintainability (See 3.11).

6.2.3.3 Calibration system (See 4.5.1.1).

6.3 Conformance rationale.

6.3.1 Reliability rationale. Reliability rationale submitted with the Bid Sample should provide clear and concise rationale showing how the reliability of the equipment complies with the minimum requirements of the solicitation. This may include data from previous reliability tests, reliability predictions and other data on the equipment being offered, such data on similar equipment or equal or greater complexity produced by the offeror may be submitted. However, such data must be clearly identified as comparative data, and accomplished with specifications and technical literature on the similar instruments. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production reliability testing.

6.3.2 Maintainability rationale. Maintainability rationale submitted with the Bid Sample should provide clear and concise rationale showing how the maintainability of the equipment complies with requirements of the solicitation. This may include data from previous maintainability tests, records of repair and calibration data, and other data on the equipment available to the offeror. Data on selection of module size and other design (See 3.4) features relevant

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to maintainability may also be submitted. The acceptance of the submitted rationale does not relieve the successful bidder of performing and successfully completing the production maintainability testing.

6.4 Supersession data. This specification supersedes MIL-O-24311(EC), MIL-O-9960C(AF), and MIL-O-83225(AF).

Custodians:
Army - EL
Navy - EC
Air Force - 99

Review Activities:
Army - MI
Navy - SH, AS

User Activities:
Army - MU, SC
Navy - OS

Preparing Activity:
Air Force - 99

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